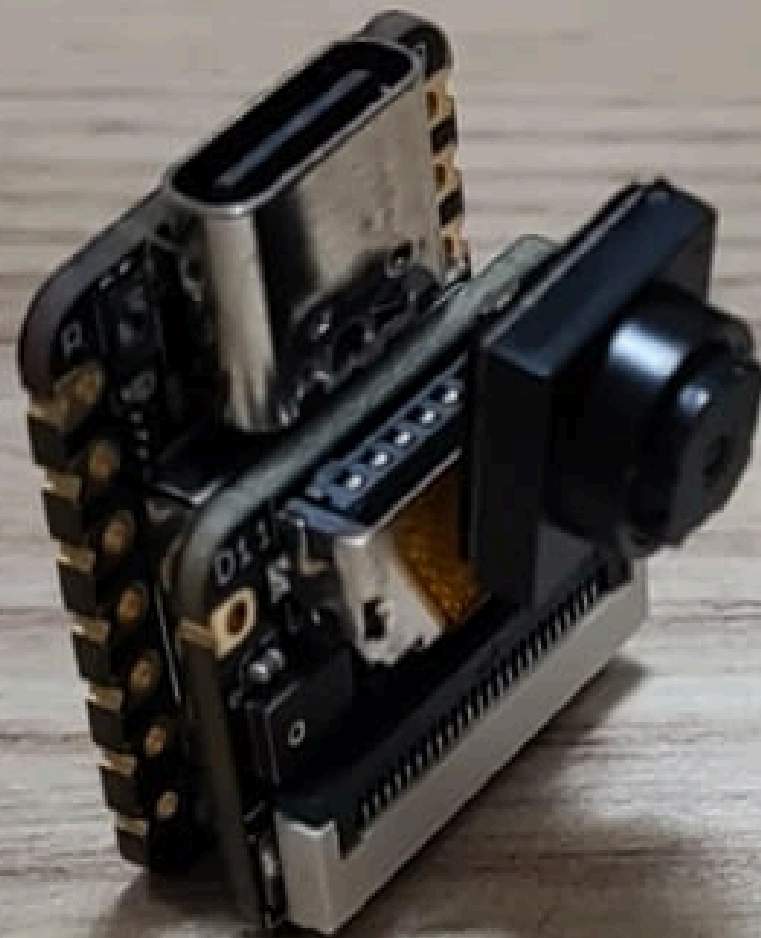


OBJECT DETECTION WITH TINYML -
CAPACITOR COUNTING

DATA
04/07/2024

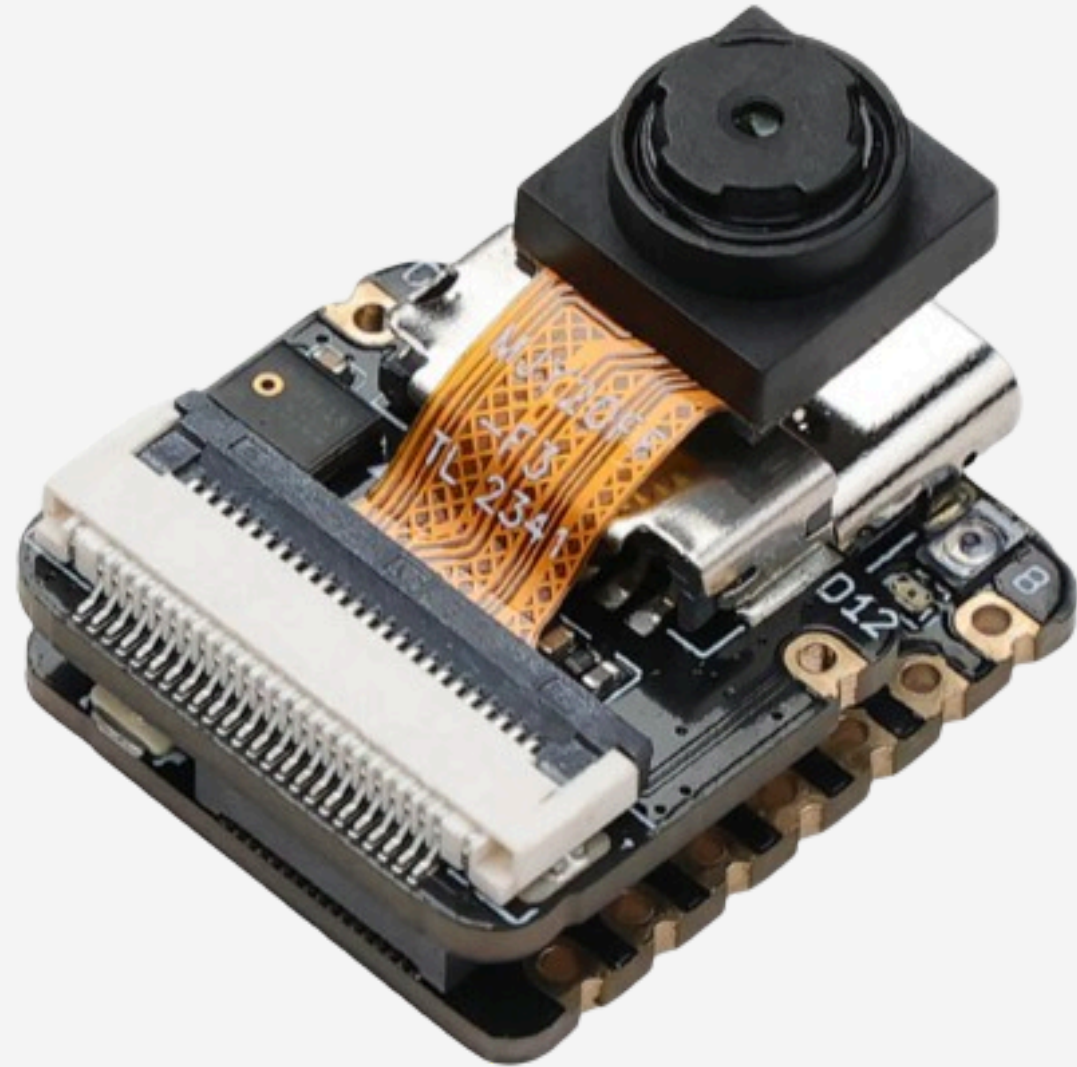


**IENTI01 - APRENDIZADO DE MAQINA
APLICADO PARA DISPOSITIVOS IOT
EMBARCADOS**

JOSÉ ANDERSON DOS REIS
SAMUEL LIMA BRAZ
TONY ALBERT LIMA

Projeto final desenvolvido na disciplina de TinyML

HARDWARE



XIAO-ESP32S3-Sense

- Seeed Studio XIAO ESP32S3 Sense
- ESP32S3 32-bit, dual-core, Xtensa processor chip operating up to 240 MHz
- 2.4GHz Wi-Fi
- BLE 5.0
- OV2640 Camera sensor
 - Resolução 1600x1200
- 8MB PSRAM e FLASH

FLUXO DE TRABALHO



01

COLETA DE DADOS

Obtenção das imagens a partir do dispositivo e armazenamento



02

PRE-PROCESSAMENTO

Limpeza de dados, Anotação dos objetos nas imagens, geração de novos dados



03

ARQUITETURA DO MODELO

Projetar e planejar a arquitetura do modelo a ser utilizado para a tarefa



04

TREINAMENTO

Treinamento do modelo selecionado e obtenção de métricas

05

ANÁLISE E OTIMIZAÇÃO

Avaliar as métricas obtidas e ajustar os hiperparâmetros do modelo

06

CONVERSÃO DO MODELO

Conversão do modelo para TFLite, processo de quantização e pruning para otimização



07

DEPLOY

Implantação e compilação do modelo no dispositivo

08

INFERÊNCIA

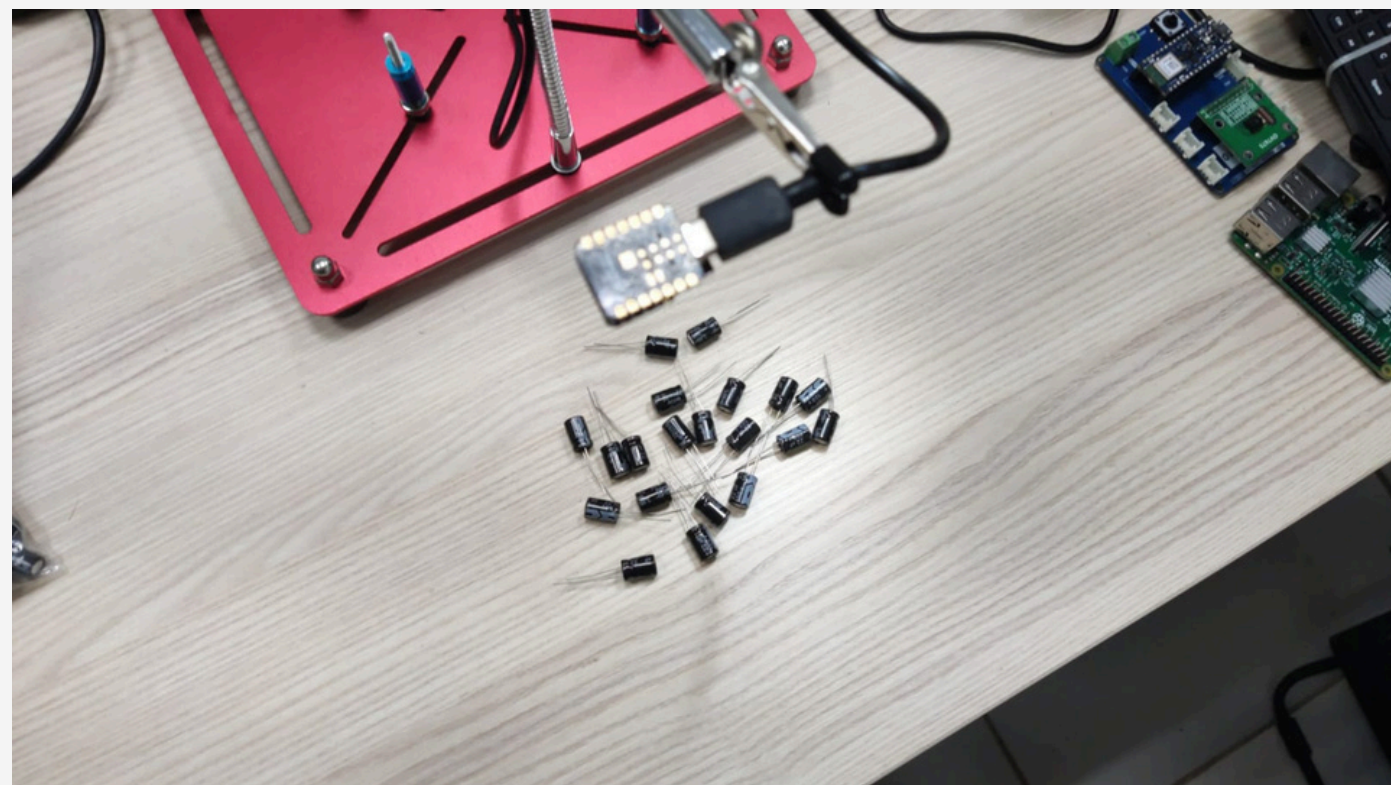
Realizar inferência do modelo em tempo real pelo dispositivo



01

COLETA DE DADOS

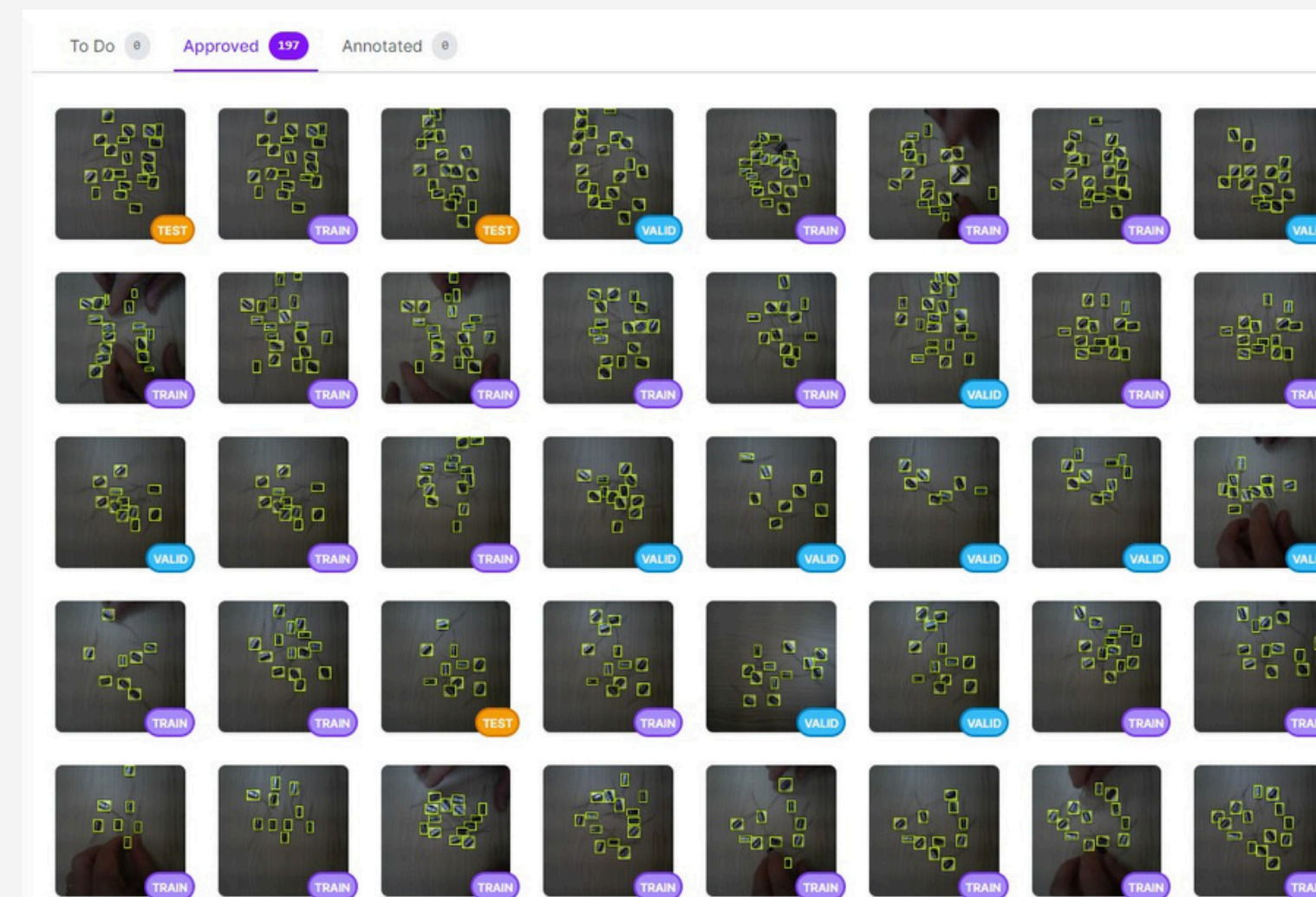
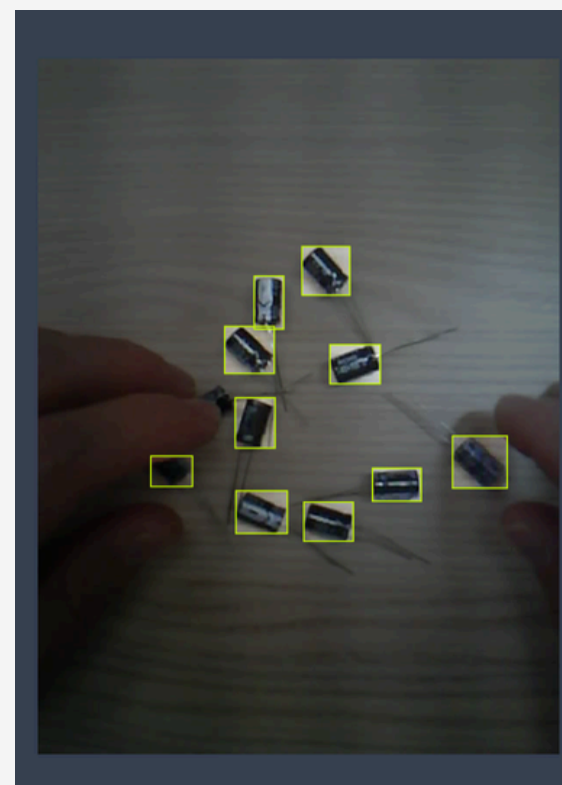
- Aquisição de dados realizada por script C++.
- Captura realizada em intervalos de tempo e armazenamento local



02 PRE-PROCESSAMENTO



- Labeling
- Data Augmentation
- Export
- [Roboflow Project](#)



606 Total Images

Preprocessing
Auto-Orient: Applied
Resize: Stretch to 640x640

Augmentations
Outputs per training example: 3
Flip: Horizontal, Vertical
Brightness: Between -25% and +25%
Exposure: Between -10% and +10%
Blur: Up to 2.5px

Export

Format

YOLOv8

TXT annotations and YAML config used with [YOLOv8](#).

☒ download zip to computer ☐ show download code

DATA COLLECTED
680 items



TRAIN / TEST SPLIT
87% / 13%

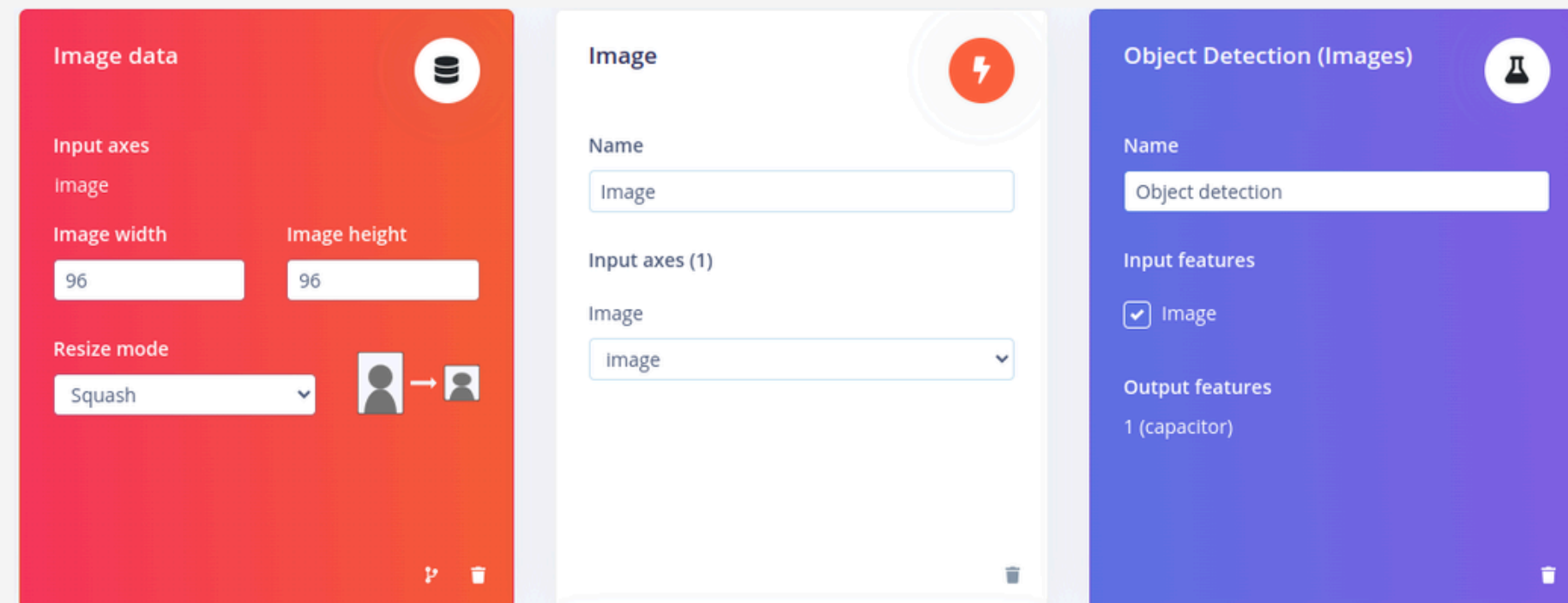


03

ARQUITETURA DO MODELO



- Impulse
 - Resize 96x96
 - Features RGB
- Object Detection
 - FOMO, identificação de centroides dos objetos
- [Edgeimpulse Studio](#)



```
mobile_net_v2 = MobileNetV2(input_shape=input_shape,
                             weights=weights,
                             alpha=alpha,
                             include_top=True)
#! Default batch norm is configured for huge networks, let's speed it up
for layer in mobile_net_v2.layers:
    if type(layer) == BatchNormalization:
        layer.momentum = 0.9
#! Cut MobileNet where it hits 1/4th input resolution; i.e. (HW/4, HW/4 C)
cut_point = mobile_net_v2.get_layer('block_3_expand_relu')
#! Now attach a small additional head on the MobileNet
model = Conv2D(filters=32, kernel_size=1, strides=1,
                activation='relu', name='head_1')(cut_point.output)
model = Conv2D(filters=32, kernel_size=1, strides=1,
                activation='relu', name='head_2')(model)
logits = Conv2D(filters=num_classes, kernel_size=1, strides=1,
                 activation=None, name='logits')(model)
return Model(inputs=mobile_net_v2.input, outputs=logits)
```

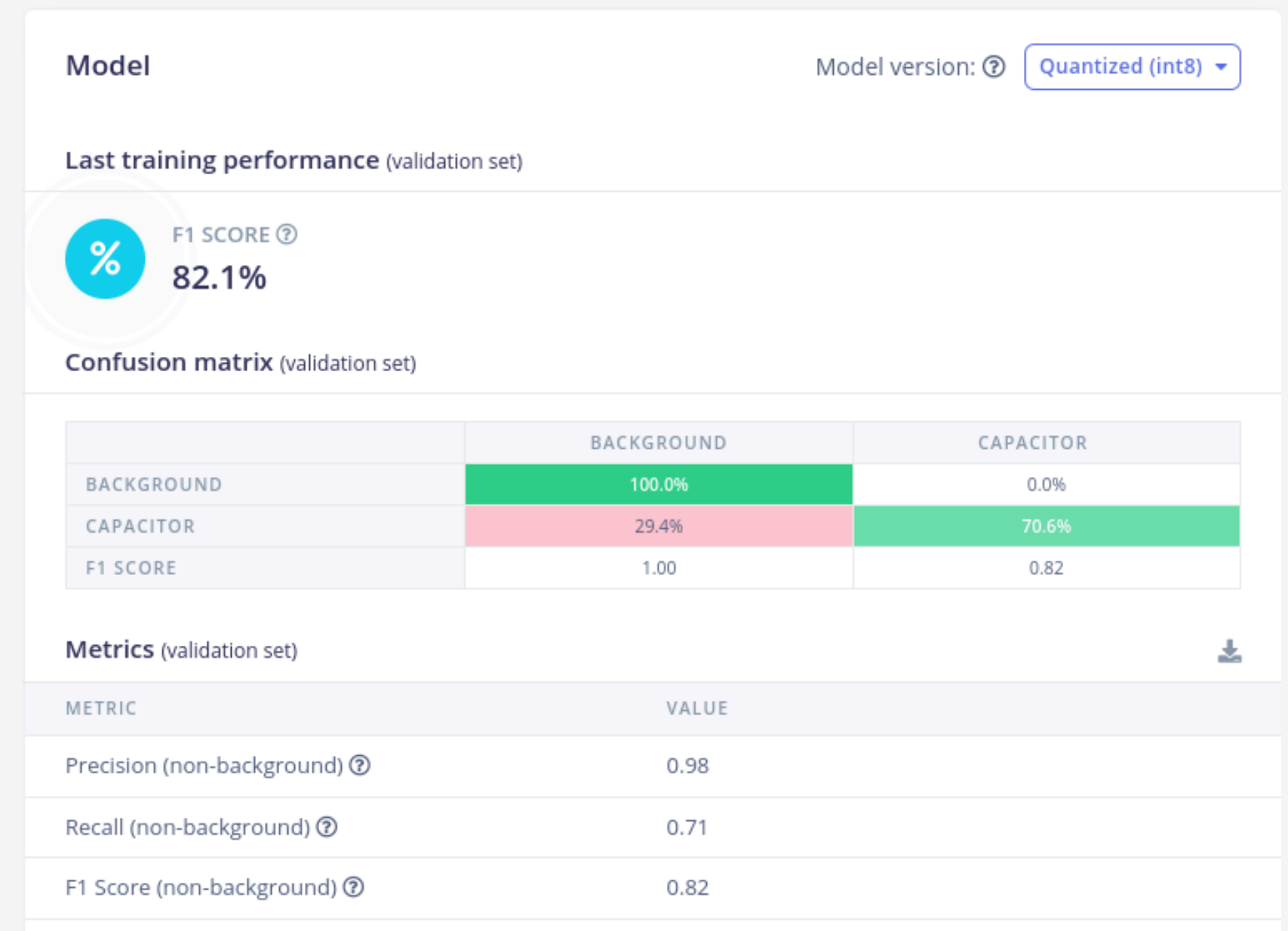
04 TREINAMENTO

- Dados de Validação (20%)
 - F1 Score: 82.1%
 - Precisão: 98%
 - Recall: 71%

Resultado adequado pelo tipo de tarefa e objeto

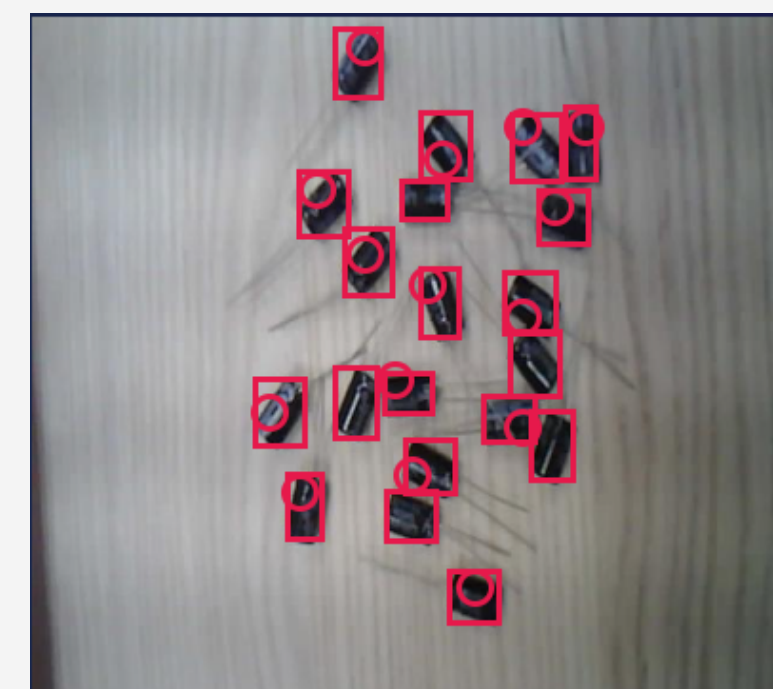
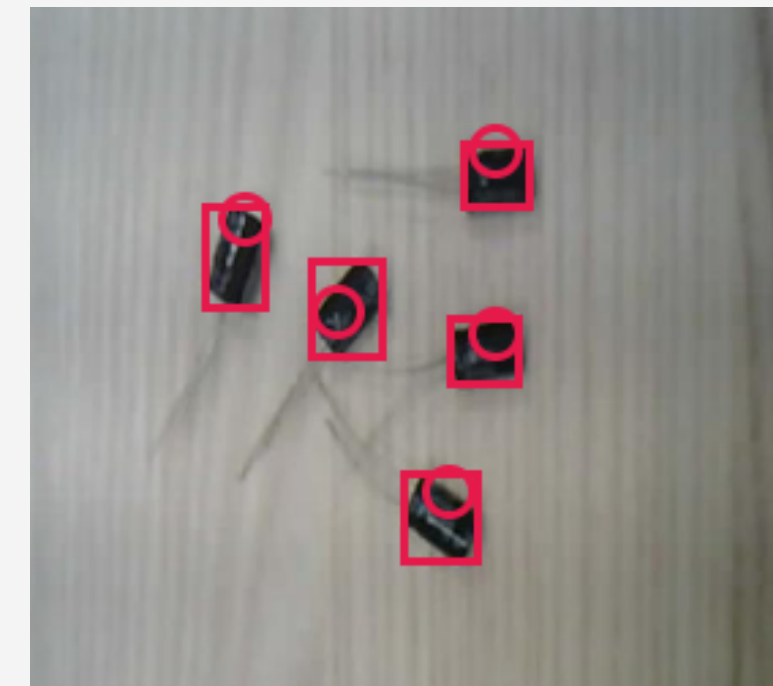
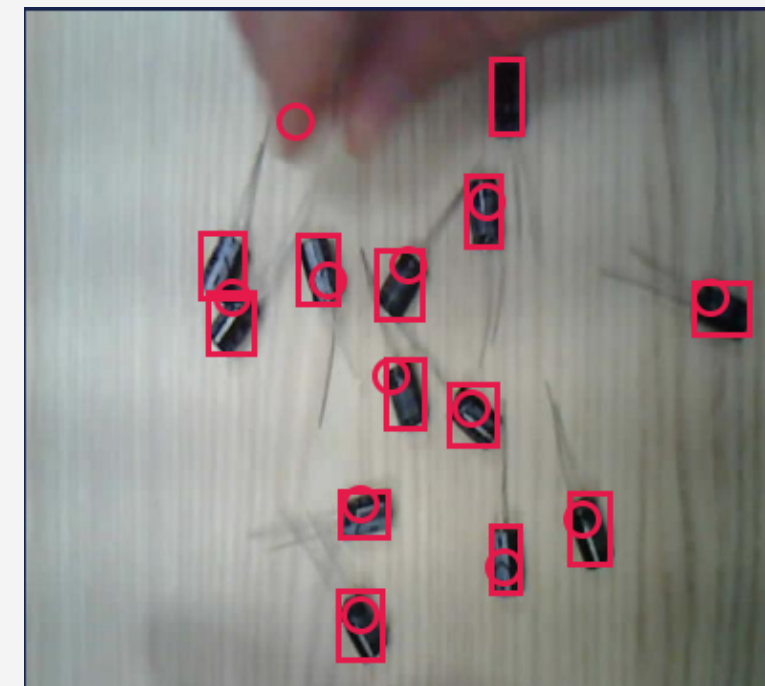
```
EPOCHS = 30
LEARNING_RATE = args.learning_rate or 0.001
BATCH_SIZE = 32

model = train(num_classes=classes,
              learning_rate=LEARNING_RATE,
              num_epochs=EPOCHS,
              alpha=0.1,
              object_weight=100,
              train_dataset=train_dataset,
              validation_dataset=validation_dataset,
              best_model_path=BEST_MODEL_PATH,
              input_shape=MODEL_INPUT_SHAPE,
              batch_size=BATCH_SIZE,
              use_volo=False,
              ensure_determinism=ensure_determinism)
```



05 ANÁLISE E OTIMIZAÇÃO

- Dados de Teste
 - Acurácia: 75.58%
 - Precisão: 97%
 - Recall: 73%
 - F1 Score: 83%



Results		Model version: ?	Quantized (int8) ▾
% ACCURACY ? 75.58%			
Metrics for Object detection ⬇			
METRIC	VALUE		
Precision (non-background) ?	0.97		
Recall (non-background) ?	0.73		
F1 Score (non-background) ?	0.83		

06

CONVERSÃO DO MODELO E DEPLOYMENT

07

- Conversão do modelo para biblioteca Arduino e compilação com EON Compiler.
- Modelo quantizado para inteiro de 8bits
- Aproximadamente 1,1 segundos para inferência e 229K de memória RAM



SELECTED DEPLOYMENT
Arduino library
An Arduino library with examples that runs on most Arm-based Arduino development boards.

MODEL OPTIMIZATIONS
Model optimizations can increase on-device performance but may reduce accuracy.



EON™ Compiler
Same accuracy, 17% less RAM, 37% less ROM.

Quantized (int8)

Selected ✓

	IMAGE	OBJECT DETECTION	TOTAL
LATENCY	15 ms.	1,164 ms.	1,179 ms.
RAM	4.0K	229.0K	229.0K
FLASH	-	54.9K	-
ACCURACY			75.58%

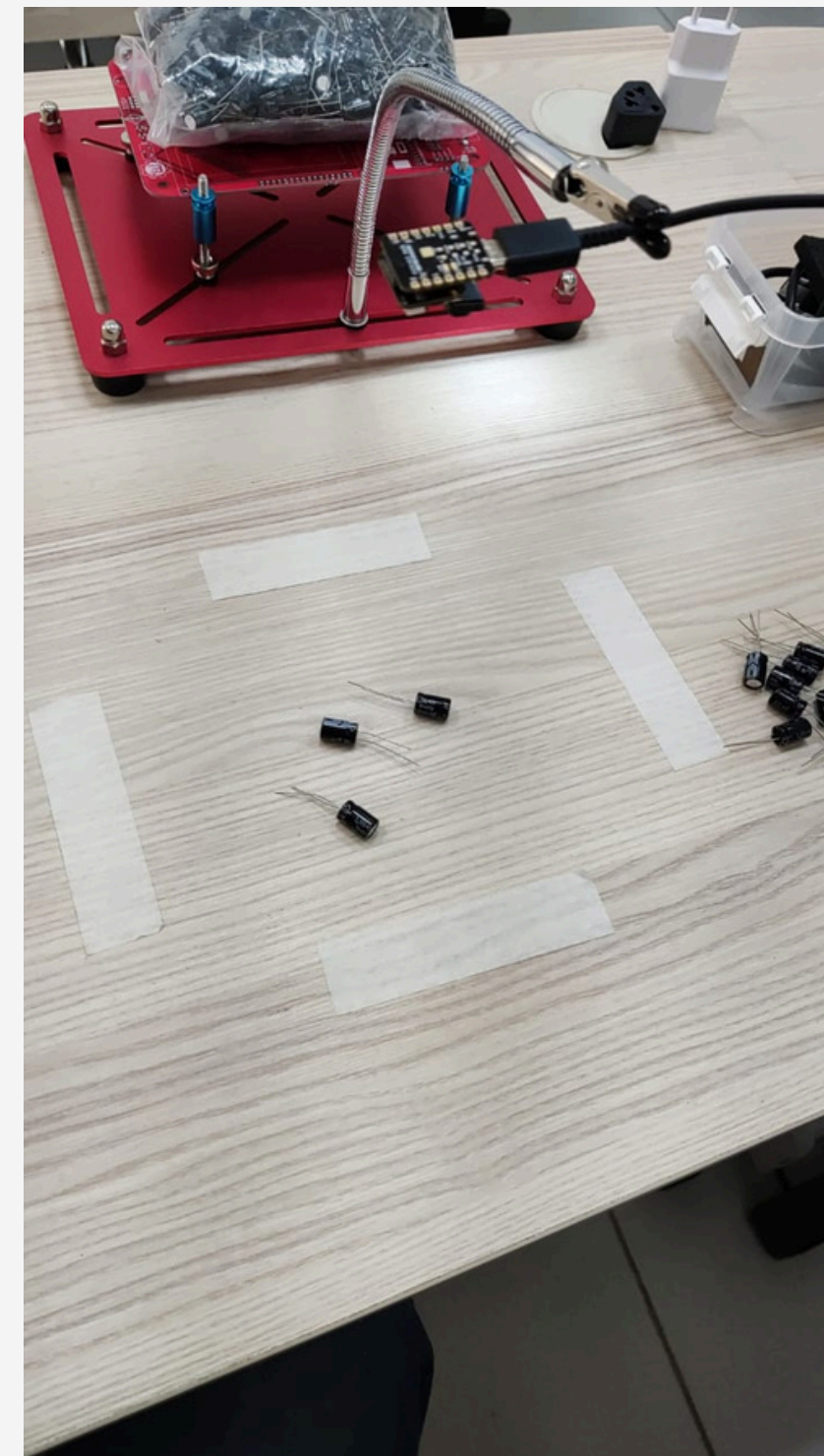
Estimate for Espressif ESP-EYE (ESP32 240MHz) - [Change target](#)

08

INFERÊNCIA

- O modelo foi corretamente compilado e carregado no Xiao permitindo a inferência local
- O processo demonstrou uma latência um pouco maior que a esperada.
- A quantidade, contagem de capacitores nos pareceu precisa em varios testes

```
Object detection bounding boxes:
  capacitor (0.984375) [ x: 8, y: 36, width: 8, height: 8 ]
  capacitor (0.757812) [ x: 32, y: 40, width: 8, height: 8 ]
  capacitor (0.972656) [ x: 36, y: 64, width: 8, height: 12 ]
Predictions (DSP: 4 ms., Classification: 3554 ms., Anomaly: 0 ms.):
Object detection bounding boxes:
  capacitor (0.976562) [ x: 8, y: 36, width: 8, height: 8 ]
  capacitor (0.667969) [ x: 32, y: 40, width: 4, height: 8 ]
  capacitor (0.964844) [ x: 36, y: 64, width: 8, height: 12 ]
Predictions (DSP: 4 ms., Classification: 3554 ms., Anomaly: 0 ms.):
Object detection bounding boxes:
  capacitor (0.988281) [ x: 8, y: 36, width: 8, height: 8 ]
  capacitor (0.757812) [ x: 32, y: 40, width: 4, height: 8 ]
```



REFERÊNCIAS

- [Rovai - object-detection-with-xiao-esp32s3-sense](#)
- [Edgeimpulse - Learning Block - Objection Detection FOMO](#)
- [Roboflow - missing-and-null-image-annotations](#)
- [Roboflow - detect-small-objects](#)
- [Seedstudio - xiao_esp32s3 - getting-started](#)
- [Github - Mjrovai -XIAO-ESP32S3-Sense](#)